

Channel Imbalance: A Unified Mechanics Account of Mental Illness

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Abstract

Current psychiatric classification describes mental illness in terms of symptom clusters — what the patient reports experiencing. Unified Mechanics offers a structural account underneath the symptoms: each major condition corresponds to a specific imbalance in the three channels of the nervous system. Anxiety is boundary channel hyperactivation. Depression is exploration channel suppression. PTSD is the boundary channel locked to a historical event. Addiction is the matter channel hijacked by artificial familiarity. ADHD is exploration channel dominance with impaired familiarity formation. OCD is the boundary unable to resolve into familiarity. Autism is high boundary sensitivity combined with deep and stable familiarity formation. Grief is the matter channel reaching for an object whose boundary no longer responds. Each condition is described in terms of its channel state, and existing treatments are reinterpreted through the same framework — not to replace clinical practice but to reveal why effective treatments work and what they are actually doing to the system.

1 Foundations

The three channels of the nervous system are described fully in the companion paper *The Three Natures in Biology*. In brief:

- **Boundary channel:** the site of crossing between states. Biologically: the synapse, the insular cortex, interoception. The felt sense of where the organism stands in relation to its environment.
- **Light channel:** signal propagating into new territory. Biologically: the action potential, the dopaminergic system, hippocampal encoding. Curiosity, novelty,

exploration.

- **Matter channel:** the settling of a state into familiarity. Biologically: long-term potentiation, the basal ganglia, myelination. Habit, automaticity, comfort.

A healthy nervous system cycles continuously through all three. A boundary is crossed (acknowledgement), a signal propagates and explores (exploration), a new familiarity forms (familiarity). The cycle then begins again from a new position.

Mental illness, in this framework, is what happens when the cycle is disrupted. One channel becomes dominant, or one channel cannot complete its function, or the handoff between channels fails. The symptoms that psychiatry observes are the downstream expression of these structural failures.

2 Anxiety: Boundary Channel Hyperactivation

In anxiety the boundary channel is running at excessive gain. The system registers threat in stimuli that do not warrant it, treating ordinary crossings as dangerous. The insular cortex reports persistent high-pressure boundary states. The autonomic nervous system remains in a state of readiness that was designed for brief crossings, not continuous occupation.

The matter channel cannot form because familiarity requires the boundary to settle. If the boundary never settles, nothing becomes familiar. The world remains perpetually uncrossed, and the organism remains perpetually at the edge.

The light channel is affected secondarily: exploration requires a degree of boundary stability to feel safe. Without it, novelty feels like threat rather than opportunity.

Why treatments work: Anxiolytics (benzodiazepines, SSRIs over time) reduce boundary channel gain — lowering the insular cortex’s sensitivity so that ordinary crossings are processed as ordinary. Vagal nerve stimulation directly reduces autonomic activation, bringing the boundary channel out of sustained alert. Meditation works by training voluntary boundary-pressure reduction: the practitioner learns to observe boundary states without amplifying them, allowing familiarity to form underneath. Exposure therapy works by forcing controlled boundary crossings in safe conditions until the system learns that the crossing resolves — and familiarity forms where fear had been.

3 Depression: Exploration Channel Suppression

In depression the exploration channel goes quiet. The dopaminergic system stops responding to novelty. What was once new produces no signal. The light channel is dim. Nothing reaches forward.

The matter channel continues to run — the person moves through familiar patterns, familiar thoughts, familiar days — but without the exploration channel to bring in new crossings, those patterns produce no reward. Familiarity without novelty is not comfort. It is hollowness. The cycle has stalled at its final stage with no way back to the beginning.

The boundary channel still operates but has nothing to cross into. A boundary that opens onto silence, again and again, produces its own particular quality of despair. The person can feel that something is wrong — the boundary is reporting — but the light channel brings no answer.

Why treatments work: Ketamine disrupts stuck matter-channel patterns directly and rapidly, forcing the system out of entrenched familiarity before the exploration channel has been restored. This is why ketamine works on treatment-resistant depression within hours rather than weeks — it breaks the deadlock rather than waiting for the exploration channel to recover. SSRIs restore baseline dopaminergic sensitivity over weeks, gradually returning the exploration channel to function. Exercise is one of the most reliable antidepressants known because it forces exploration channel activation through physical novelty and exertion — the body exploring its own capacity. Psychedelics produce massive, controlled exploration channel activation, introducing crossings so numerous and unfamiliar that the stuck matter patterns are disrupted and new familiarity must form. The therapeutic window after a psychedelic session is the period in which the system is open to forming new familiarity rather than defaulting to old patterns.

4 PTSD: The Boundary Locked to a Historical Crossing

In PTSD a boundary crossing of overwhelming intensity has failed to resolve. The event was too large for the available familiarity to absorb. The boundary opened, the system could not close it, and it has remained open ever since — not in the present but replayed from the past, as if the crossing is still occurring.

The body does not distinguish between a real boundary crossing and a replayed one. The insular cortex fires the same signals. The autonomic system responds with the same urgency. The organism is locked at the moment of an event that has already ended, returning to it whenever the boundary channel is triggered by anything sufficiently similar to the original crossing.

The matter channel cannot form around the event because every approach to it reopens the boundary before familiarity can settle. The person cannot move through the crossing because the crossing will not complete.

Why treatments work: EMDR (eye movement desensitisation and reprocessing) works by activating the boundary crossing in a controlled setting while the bilateral stimulation reduces the autonomic gain on the event. The crossing is re-entered at lower intensity, again and again, until the system is able to move through it and familiarity forms on the other side. Prolonged exposure therapy operates on the same principle: systematic re-crossing of the boundary under safe conditions until the event is no longer experienced as present. Propranolol, administered during memory reconsolidation, reduces the autonomic intensity of the replay, making it possible for the boundary to settle. All effective PTSD treatments are functionally identical: they find a way to let the boundary crossing complete so that familiarity can finally form.

5 Addiction: The Matter Channel Hijacked

The matter channel is designed to be reached through the full cycle: boundary crossed, exploration undertaken, familiarity earned. This sequence is what makes natural rewards — food, connection, achievement, rest — feel satisfying. The familiarity is real because the crossing was real.

Addictive substances short-circuit this cycle. They deliver the matter channel's reward — the feeling of familiarity and resolution — without requiring the boundary to be crossed or the exploration to occur. Dopamine floods the system on demand. Familiarity arrives without being earned.

The consequence is that the natural cycle atrophies. The boundary crossings of ordinary life produce less and less reward because the matter channel has been calibrated to the artificial intensity. Nothing that requires the full cycle feels worth doing. The substance is the only thing that reaches the matter channel at all. Familiarity has been hijacked: the organism is now deeply familiar with the substance and increasingly unfamiliar with everything else.

Why treatments work: Cold cessation is difficult because it removes the only functioning path to the matter channel while the natural cycle has not yet recovered. Replacement therapies (methadone, nicotine replacement) maintain minimal matter-channel access while the natural cycle is slowly restored. The work of recovery is the work of re-learning how to cross boundaries, explore, and earn familiarity through the full sequence — which is why connection, purpose, and structured activity are consistently the most effective long-term supports. They restore the cycle rather than substituting for it.

6 ADHD: Exploration Dominant, Familiarity Impaired

In ADHD the exploration channel is dominant and the threshold for novelty is low. What is familiar becomes flat very quickly because the light channel moves on before the matter channel has time to consolidate. The person is not deficient in dopamine — they are more sensitive to the absence of novelty. The exploration channel demands constant new crossings. Anything already familiar produces no signal, so attention moves elsewhere.

This is not a failure of will. It is a channel balance in which the light channel outpaces the matter channel's ability to form. The cycle runs faster than it can settle.

Why treatments work: Stimulants are paradoxical: they calm ADHD rather than stimulating it further. The reason is that they normalise the dopaminergic signal rather than amplifying it. By raising baseline dopamine, they reduce the threshold gap between familiar and novel, allowing the matter channel to hold a state long enough to become genuinely familiar. The exploration channel no longer needs to move on so quickly because the current state feels sufficiently rewarding. Structure and routine serve the same purpose: they artificially extend the familiarity phase, giving the matter channel time to consolidate what the exploration channel has found.

7 OCD: The Boundary That Cannot Resolve

In OCD the boundary crosses but familiarity does not form. The system expects resolution after the crossing and does not receive it. So it crosses again. And again. Not because the person wishes to but because the channel logic demands that a boundary crossing produces a settling, and when the settling does not come the only available response is to try again.

The compulsion is not the disorder. The compulsion is the system's attempt to solve the disorder. It is the matter channel being reached for through an action that used to resolve the boundary but no longer does. The boundary keeps reopening before familiarity can stabilise.

Why treatments work: Exposure and response prevention (ERP) is the most effective treatment for OCD. It works by forcing the system to sit with the open boundary without performing the compulsion. The discomfort is intense because the boundary channel is reporting an unresolved state and demanding action. But if the compulsion is withheld, something happens that the system had stopped believing was possible: the boundary eventually settles on its own. Familiarity forms. The cycle completes without the compulsion. Each successful exposure teaches the system

that resolution is available without the ritual — and gradually the boundary learns to settle.

8 Autism: High Boundary Sensitivity, Deep Familiarity

Autism is not a deficit. It is a specific channel balance: the boundary channel is highly sensitive, and the matter channel forms deep, stable, and reliable familiarity. These two features are not independent — they are aspects of the same underlying calibration.

High boundary sensitivity means that crossings which are unremarkable to most nervous systems register with full force. Sensory experience is intense because the boundary channel processes all input at high gain. Social situations are demanding because they involve continuous rapid boundary crossings with unpredictable timing. The world is always somewhat louder than expected.

But the matter channel, once it has formed familiarity, holds it with exceptional stability. Interests become deep rather than broad. Routines provide genuine relief because they are the matter channel operating in conditions of low boundary pressure. The familiar is not boring — it is genuinely satisfying in a way that shallow familiarity is not.

What is labelled as rigidity is the matter channel protecting itself from boundary overload. What is labelled as special interest is the light channel going very deep in a direction where the boundary pressure is manageable. Neither is pathological. They are the predictable outputs of a nervous system with this particular channel balance.

9 Grief: Familiarity Without a Boundary

Grief is a channel state without a clinical category adequate to describe it, because it is not an illness. It is the matter channel working exactly as it should, in a situation that has become impossible.

When a person loses someone they love, the matter channel has formed deep familiarity with that person — with their presence, their voice, their particular way of being in the world. That familiarity does not end with the loss. The matter channel continues to reach for it. It expects the boundary to respond, as it always has, and receives silence.

Grief is that silence. The matter channel running the familiarity pattern and finding

nothing to cross into. The boundary where someone used to be.

This is why grief is physical. Why it arrives in waves rather than steadily. Why it can be triggered by unexpected things — a song, a smell, a particular quality of afternoon light. Each trigger activates the familiarity pattern and the boundary reaches forward and finds nothing.

Grief does not end when the person stops thinking about the loss. It ends — partially, gradually, in patches — when new familiarity forms that does not require the absent boundary to respond. The lost person becomes part of the matter channel in a different way: no longer as a present crossing, but as part of the accumulated weight of what has made the system what it is.

10 Summary Table

Condition	Channel State	Effective Treatment Logic
Anxiety	Boundary hyperactivated	Reduce boundary gain; allow familiarity
Depression	Exploration suppressed	Force exploration channel back online
PTSD	Boundary locked to past event	Complete the unfinished crossing
Addiction	Matter channel hijacked	Restore the full natural cycle
ADHD	Exploration dominant	Normalise novelty threshold
OCD	Boundary cannot resolve	Let boundary settle without compulsion
Autism	High boundary sensitivity	Accommodate; not a disorder to fix
Grief	Matter reaching for absent boundary	Time; new familiarity forming

The standard model of mental illness treats conditions as categories of symptom. This framework treats them as states of a system. The difference matters clinically: a system account predicts which treatments will work and why, suggests what to try when first-line treatments fail, and identifies the underlying mechanism rather than managing the surface expression.

A chemical imbalance is real. But it is the downstream expression of a channel imbalance. The chemistry is how the channel state is implemented in tissue. The channel state is what needs to be understood.